

Programming Contest for Girls

Example Problems v3.1

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Days in a Leap Year

Problem 1 (5 points)

Print, as a single integer on a single line, how many days there are in a leap year.

Hints

- Use `print()` to output your answer.
- There is no penalty for making guesses.

Limited Addition

Problem 2 (5 points)

You and your partner have spotted a limited-addition Squishmallow[1] which you must have. Quick! How much money can you scrape together? As input, you will receive two positive **integers** each on their own line, representing the money you each contribute. Print the total amount you have altogether.

Input 1

```
3
5
```

Output 1

```
8
```

Input 2

```
4
7
```

Output 2

```
11
```

Hints

- Use `int(input())` to read a line of text into your program and convert it to an integer.
- **Do not** use a prompt like `input("Person 1: ")` as these will cause your submission to be incorrect.

Stickers

Problem 3 (5 points)

Your job at the local store requires you to write stickers advertising an item on special. You're given the name of the item that is on special, followed by its special price, each on its own line.

Print out "`X are on special for $Y!`" where `X` is replaced by the name of the on-special item and `Y` is replaced by the special price.

Input 1

```
Fireworks
29.99
```

Output 1

```
Fireworks are on special for $29.99!
```

Input 2

```
Lemon tea bags
5
```

Output 2

```
Lemon tea bags are on special for $5!
```

Hints

- Take care to match the punctuation, capitalisation, and spacing **exactly** for the output.
- Check the cheat sheet if you're stuck.

[1] Squishmallows are collectable plush toys.

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Tech Support

Problem 4 (5 points)

Your friend is always asking for help using her phone. She asks the same questions so frequently that you've decided to write a program to automate your replies. As input, you will receive two space-separated strings on a single line, representing the the words to insert into your reply. Print the phrase "`Don't worry! Go to X and then find Y.`", where `X` is replaced with the first given string and `Y` is replaced with the second given string.

Input 1

```
Settings Bluetooth
```

Output 1

```
Don't worry! Go to Settings and then find Bluetooth.
```

Hint

- Try using `.split()` to divide an input line into a list of strings.
- Take care to match spelling and punctuation exactly.

AI Chat-bot

Problem 5 (5 points)

AI chat-bots work by predicting which token is most likely to come next. Your AI chat-bot can only predict one of two possible tokens depending on the last character of input. Input will be a string of at-least one character on a single line. If the last character is "`?`" then print "`Yes.`". Otherwise, print "`Okay.`".

Your AI is rather agreeable.

Input 1

```
Would I look good in a bucket hat?
```

Output 1

```
Yes.
```

Input 2

```
When I'm Prime Minister, I will make avocadoes free for all to enjoy.
```

Output 2

```
Okay.
```


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Ribbons

Problem 6 (5 points)

You and your friend are making decorations for a birthday party using ribbon. Write a program that calculates how many decorations you can make with a given length of ribbon.

The first line of input is a single non-negative integer, the length of ribbon you have. The next line contains the length, as a non-negative integer, of ribbon you need to make a single decoration.

Output the number of decorations you can make with the ribbon you have on a single line. On the next line, output the amount of ribbon you will have left over.

Input 1

```
5
3
```

Output 1

```
1 2
```

Input 2

```
30
10
```

Output 2

```
3 0
```

Hint

- What mathematical operations could help here?

Alex's Apples

Problem 7 (10 points)

By Georgia

This problem has two parts. Submit each part to receive points.

Part 1

Alex owns a small apple orchard in Hawkes Bay. She wants to deliver these apples to the nearest supermarket. First, Alex must pack the apples into special boxes to ensure they don't get bruised or damaged. Unfortunately, these high tech safe boxes cost a lot of money. If Alex gives you the number of apples to be delivered, the capacity of each box (in apples), and the number of boxes they can afford, can you tell them if they're able to deliver all the apples safely?

You will be given the number of apples grown that month, the maximum number of apples that can fit in one box, and the number of boxes they can afford that month, as positive integers on three individual lines.

If every apple can be packed into a box, output "`All apples safely stored.`".

Otherwise, output "`Apples not safely stored.`"

Input 1

```
8
3
3
```

Output 1

```
All apples safely stored.
```

Input 2

```
50
10
4
```

Output 2

```
Apples not safely stored.
```

Part2

The scenario is the same, except the output should be as follows instead...

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- If every apple can be packed into a box output "`All apples safely stored.`".
- If exactly one apple cannot be stored, output "`An apple is unsafe.`".
- If multiple apples cannot be stored, output "`'X' apples are unsafe.`", where `X` is the number of apples that cannot be stored.

Input 1
```
7
2
3
```

Output 1
```
An apple is unsafe.
```

Input 2
```
8
3
3
```

Output 2
```
All apples safely stored.
```

Input 3
```
50
10
4
```

Output 3
```
'10' apples are unsafe.
```

Spill the Tea

Problem 8 (10 points)

Social media evolves so quickly that the latest in-phrase changes every day. You decide to make a program that automatically reminds you of the latest in-phrase each morning.

The first line of input contains a single integer, representing the total number of days. For each day, the in-phrase appears on a single line.

For each day, print "`Day X: The phrase today is "".`" with the phrase inside the double quotes and the day number in place of `X`. Make sure to double-check punctuation and formating!

Input 1
```
3
OMG
No cap
Skibidi
```

Output 1
```
Day 1: The phrase today is "OMG".
Day 2: The phrase today is "No cap".
Day 3: The phrase today is "Skibidi".
```

Supermarket Search

Problem 9 (10 points)

Inspired by Sophie

You are scanning the shelves of a supermarket looking for honey to sweeten your lemon tea. You have a list of labels from the supermarket shelf, finishing with the label "`END`". If the label "`Honey`" appears at-least once then print "`Sweet tea, full of glee.`". Otherwise, print "`Sour tea, it cannot be!`"

Input 1
```
Avocados
Honey
Toilet Paper
END
```

Output 1
```
Sweet tea, full of glee.
```

Input 2
```
Candles
Insecticides
Hopes & Dreams
END
```

Output 2
```
Sour tea, it cannot be!
```


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Wardrobe

Problem 10 (20 points)

Every day, Charlie adds a new item to her wardrobe. Sometimes this is a spontaneous op-shop buy and other times it's a piece she has been saving towards for months. Charlie has promised herself to stop once she acquires an 'ideal wardrobe'. You will be given a space separated list of unique names describing the items in Charlie's ideal wardrobe. The following lines each contain the item Charlie gets that day. The last line will be "`STOP`". Output the day number (counting from 1) on which Charlie gets the item that completes her wardrobe. If she never completes her wardrobe, output "`It be like that.`". Only count exact matches.

Input 1

```
green turquoise
turquoise
blue
STOP
```

Output 1

```
It be like that.
```

Input 2

```
orange lilac black
green
lilac
orange
beige
black
white
STOP
```

Output 2

```
5
```

Party Hats

Problem 11 (20 points)

Write a program to draw a party hat of a given size. A party hat forms an equilateral triangle with a given number of * on each side. There is a single space between each star on the last row. There is no leading space on the last row. A hat is symmetric about its vertical centreline. A hat of size 1 is just a single star. Input is a single positive integer (greater than zero) on a single line, the number of asterisks along each side. Your output must match the expected output exactly.

Input 1

```
3
```

Output 1

```
  *
 * *
* * *
```

Input 2

```
4
```

Output 2

```
   *
  * *
 *   *
* * * *
```


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